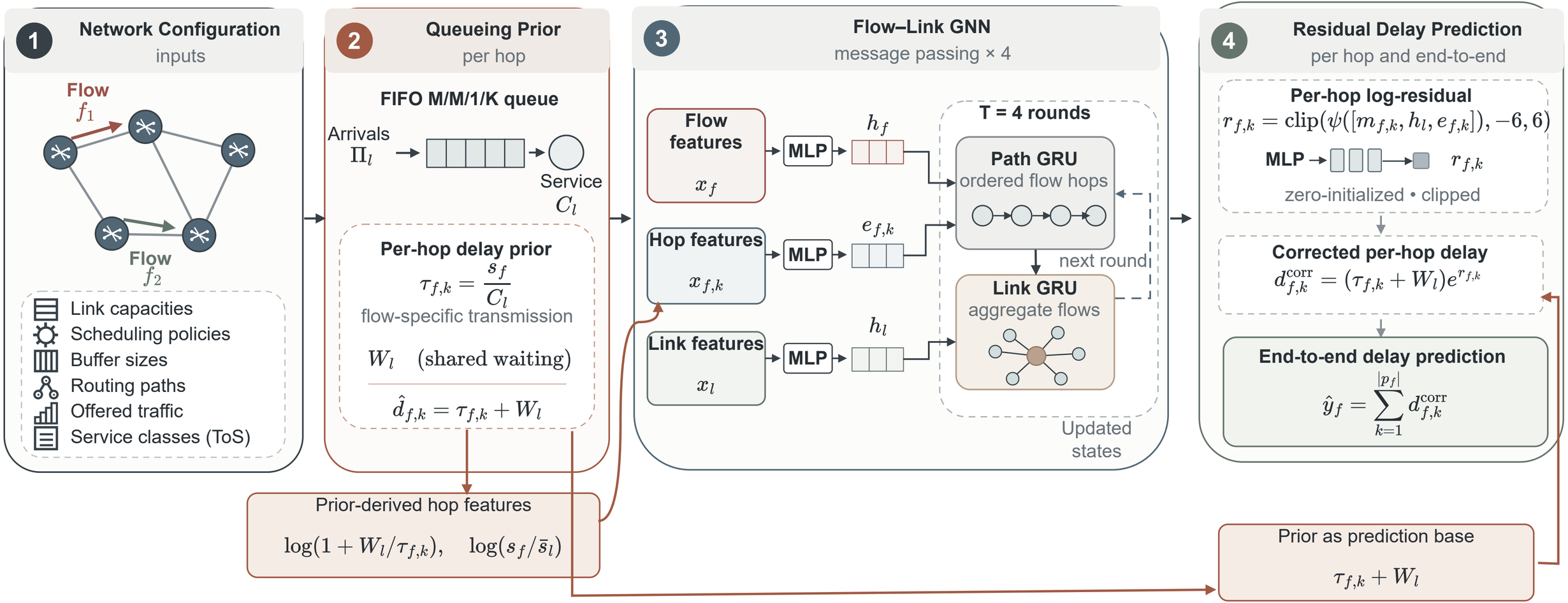

1
Network Configuration
inputs
Flow
f_1
Flow
f_2
Link capacities
Scheduling policies
Buffer sizes
Routing paths
Offered traffic
Service classes (ToS)
2
Queueing Prior
per hop
FIFO M/M/1/K queue
Arrivals
Π_l
Service
C_l
Per-hop delay prior
$\tau_{f,k} = \frac{s_f}{C_l}$
flow-specific transmission
W_l (shared waiting)
$\hat{d}_{f,k} = \tau_{f,k} + W_l$
Prior-derived hop features
$\log(1 + W_l/\tau_{f,k}), \quad \log(s_f/\bar{s}_l)$
3
Flow–Link GNN
message passing × 4
Flow features
x_f
Hop features
$x_{f,k}$
Link features
x_l
MLP
MLP
MLP
h_f
$e_{f,k}$
h_l
T = 4 rounds
Path GRU
ordered flow hops
next round
Link GRU
aggregate flows
Updated states
4
Residual Delay Prediction
per hop and end-to-end
Per-hop log-residual
$r_{f,k} = \mathrm{clip}(\psi([m_{f,k}, h_l, e_{f,k}]), -6, 6)$
MLP
$r_{f,k}$
zero-initialized • clipped
Corrected per-hop delay
$d^{\mathrm{corr}}_{f,k} = (\tau_{f,k} + W_l)e^{r_{f,k}}$
End-to-end delay prediction
$\hat{y}_f = \sum_{k=1}^{|p_f|} d^{\mathrm{corr}}_{f,k}$
Prior as prediction base
$\tau_{f,k} + W_l$